

DATA SCIENCE — CAT — PHASE II — SESSION PLAN

CAT I vs CAT II — Program Positioning

Aspect	CAT 1 (Phase 1)	CAT 2 (Phase 2)
Work mode	Individual / self-preparation	Team-based
Focus areas	Level 1 (ML, DL), Level 2 (NLP, SLM), Level 3 (Gen AI, Agentic AI)	Team should prepare for the levels based on their assigned roles
Goal	Build individual conceptual and technical foundation across the AI stack	Demonstrate ability to integrate that knowledge into a working, collaborative project
Deliverable	Likely individual assignments, mini-implementations, or topic-wise submissions per domain (ML/DL/NLP/etc.)	A single team project — likely an end-to-end system (e.g., combining NLP/Gen AI/Agentic AI into one application)
Skills tested	Conceptual clarity, self-driven learning, ability to implement standalone models/techniques	Collaboration, project planning, integration of multiple components, division of labor
Evaluation criteria (typical)	Correctness/depth of individual topic coverage, originality of implementation, understanding demonstrated	Team coordination, project scope/complexity, functional end-to-end system, individual contribution within the team, presentation/demo quality
Risk factors	Time management across many topics since it's solo	Coordination overhead, uneven contribution, integration bugs between team members' parts

Day-Wise Plan

Every team stays with its assigned client problem throughout. Roles are fixed from Day 1. Each non-checkpoint day opens with a Project Implementation Guide session instead of a lecture.

DAY 1 · INTRODUCTION

FN — Forenoon	AN — Afternoon
• One-time client-problem briefing to all teams (Group A → Problem 1, Group B → Problem 2). • One-time role assignment — each student is fixed into one functional role for the entire program (no rotation). • Application demo	• Project Implementation Guide — Machine Learning

DAY 2 · MACHINE LEARNING (ML)

FN — Forenoon	AN — Afternoon
• Project Implementation (ML)	• Debate • Project Implementation (ML)

DAY 3 · DEEP LEARNING (DL)

FN — Forenoon	AN — Afternoon
• Project Implementation Guide — Deep Learning • Project Implementation (DL)	• Debate • Project Implementation (DL)

DAY 4 · EVALUATION 1

FN — Forenoon	AN — Afternoon
• Project Machine Learning Evaluation	• Project Deep Learning Evaluation

DAY 5 · NLP

FN — Forenoon	AN — Afternoon
• Project Implementation Guide (NLP) • Project Implementation (NLP)	• Debate • Project Implementation (NLP)

WEEK 1 MILESTONE · Day 1 – Day 5

By the end of Week 1, every team has moved from client-problem briefing and fixed role assignment through a complete Machine Learning build-evaluate cycle and a Deep Learning build-evaluate cycle, and has kicked off NLP. Teams have learned to collect and clean data, run exploratory analysis, build and tune models, and evaluate them with proper metrics — for both ML and DL. Week 1 closes with the NLP text pipeline (corpus collection, tokenization, and an initial classification/NER/sentiment model) actively in progress.

DAY 6 · SLM	
FN — Forenoon	AN — Afternoon
• Project Implementation Guide — SLM • Project Implementation (SLM)	• Debate • Project Implementation (SLM)

DAY 7 · EVALUATION 2	
FN — Forenoon	AN — Afternoon
• Project Evaluation — NLP	• Project Evaluation — SLM

DAY 8 · GEN AI	
FN — Forenoon	AN — Afternoon
• Project Implementation Guide — Gen AI • Project Implementation (Gen AI)	• Debate • Project Implementation (Gen AI)

DAY 9 · AGENTIC AI	
FN — Forenoon	AN — Afternoon
• Project Implementation Guide — Agentic AI • Project Implementation (Agentic AI)	• Debate • Project Implementation (Agentic AI)

DAY 10 · EVALUATION 3	
FN — Forenoon	AN — Afternoon
• Project Evaluation — Gen AI & Agentic AI	• Final Capstone Quizmania

WEEK 2 MILESTONE · Day 6 – Day 10
By the end of Week 2, every team has carried that same build-evaluate discipline through Small Language Models, Generative AI, and Agentic AI — fine-tuning and benchmarking an SLM, designing prompts and building a GenAI generation pipeline, and architecting an agent with reasoning, planning, and tool-calling logic. Each stage was evaluated against its own metrics and integrated into the team's application. Week 2 closes with the Final Capstone Quizmania.

Fixed Roles — Same 5 Roles

Every team keeps the same 5-person role split across all 6 sections. The role name below is what each student owns, section after section — only the underlying subject matter changes (ML → DL → NLP → SLM → Gen AI → Agentic AI).

Machine Learning

Role	Responsibility
Data Engineer	Collect, clean, and prepare structured/tabular data for the ML pipeline
EDA Engineer	Perform exploratory data analysis — distributions, correlations, feature insights
ML Engineer	Build, train, and tune machine learning models
Evaluation Engineer	Evaluate model performance using appropriate metrics; error analysis
Integration Engineer	Integrate the ML model into an application/API

Deep Learning

Role	Responsibility
Data Engineer	Collect and preprocess image/audio/tabular data; perform augmentation
EDA Engineer	Analyze data distribution and feature/pixel-level insights
DL Engineer	Design model architecture (CNN/RNN/Transformer) and train the model
Evaluation Engineer	Assess accuracy, loss curves, confusion matrix; tune hyperparameters
Integration Engineer	Integrate the trained DL model into an application/API

NLP

Role	Responsibility
Data Engineer	Collect, clean, and label text corpus
EDA Engineer	Analyze text statistics, word frequency, embedding patterns
NLP Engineer	Handle tokenization and build the NLP model (classification/NER/sentiment etc.)
Evaluation Engineer	Evaluate using precision/recall/F1-score; perform error analysis
Integration Engineer	Integrate NLP model into chatbot/API/UI

SLM (Small Language Model)

Role	Responsibility
Data Engineer	Curate and format datasets for fine-tuning the small language model
EDA Engineer	Check dataset quality and analyze token distribution
SLM Engineer	Fine-tune and train the small language model
Evaluation Engineer	Benchmark using perplexity, task accuracy, and latency metrics
Integration Engineer	Deploy the SLM on a lightweight/edge application

Gen AI

Role	Responsibility
Data Engineer	Collect and clean training/reference datasets
EDA / Prompt Engineer	Analyze data and design/experiment with prompts
GenAI Engineer	Build the generation model/pipeline (text/image/audio)
Evaluation Engineer	Assess output quality (BLEU/FID/human evaluation)
Integration Engineer	Integrate the GenAI pipeline into a user-facing application

Agentic AI

Role	Responsibility
Data/Knowledge Engineer	Build knowledge base and set up tools/APIs for the agent
Workflow Engineer	Break down tasks and design agent workflow & tool-mapping
Agent Engineer	Design agent architecture, reasoning/planning, and tool-calling logic
Evaluation Engineer	Measure task success rate, reasoning accuracy; analyze failures
Integration Engineer	Deploy the agent and build the orchestration UI